

Thermal history sensor based on glass-ceramics

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Abstract

The degradation of mechanical properties of materials during service is strongly dependent on the history of thermal exposure resulting from thermally activated microstructural processes. This degradation can lead to catastrophic failure of engineering components; consequently, knowledge of the thermal history of a component is crucial to predicting the degradation of properties and anticipating/preventing failures. In this work, a thermal history sensor capable of operating wirelessly in severe environments has been developed which functions as much more than a thermal fuse, but less than a computer-monitored thermocouple. The sensor utilizes the thermally activated crystallization of glass-ceramics to record a thermal history fingerprint in an array of glass-ceramic substrates. Computer modeling as well as experimental results are presented as proof of concept for the sensor. Degradation state sensing as well as the effects of environmental moisture on the operation of the sensor will also be discussed. The end product will employ a pattern-matching algorithm to characterize instantly the thermal exposure by comparing to a database of thermal history fingerprints. The thermal history sensor is expected to be useful in a variety of high temperature, severe environment applications. © 2007 Elsevier B.V. All rights reserved.

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1. Introduction

The degradation of materials and engineering components in service directly correlates with the mechanical loads, thermal cycles, environmental exposure, etc., experienced during service. In this sense, the material/component “knows” its history of exposure and the degradation is itself a record of such exposures. The ability to extract this history of exposure from the material/component often requires a destructive test that characterizes the degradation state. Any method to obtain the same information non-destructively will help in predicting the remaining useful lifetime of the material or component as well as in anticipating/preventing component failures. These capabilities allow for improvements in safety as well as potential savings by preventing premature replacement or maintenance of engineering components.

In the case of thermal exposure, degradation of materials arises due to thermally activated microstructural phenomena such as grain growth, precipitate coarsening, diffusion, and

oxidation. In such processes, the temperature dependence of the rate constants typically follows an Arrhenius-type relation, which predicts a strong (exponential) dependence on temperature combined with a moderate (linear or parabolic) dependence in time. The combination of time and temperature determines the total extent of degradation. Such time- and temperature-dependent microstructural changes often lead to diminished mechanical properties and subsequently to failure of the components. Consequently, knowing the history of thermal exposure, that is, the time-dependent temperature profile, for materials/components susceptible to such phenomena is essential to diagnosing unhealthy components and preventing catastrophic failures.

In the current work, a novel thermal history sensor is described which provides more relevant information and is easier to use than either thermal fuses or monitored thermocouples. A conceptual schematic of the operation of the sensor is shown in Fig. 1. The sensor is envisioned to be small and unobtrusive with dimensions similar to a small coin; a small sensor size will assure rapid heat transfer to the sensor and improve the sensitivity of the device. The sensor is fastened to a component in such a way as to be forced to undergo the same thermal exposure as the component; following exposure, at ambient conditions,

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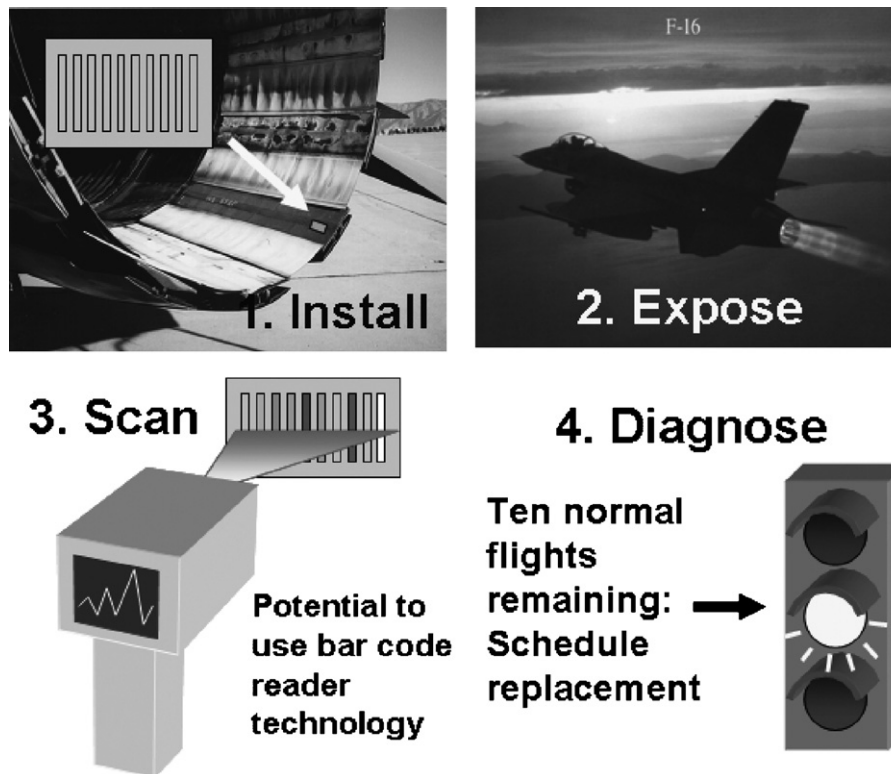


Fig. 1. Conceptual illustration of sensor operation.

the sensor is scanned or read to extract a signal that contains information regarding the time and temperature of exposure. By extracting the thermal history in this manner, the health of the component following exposure may be ascertained. It is shown that the sensor is able to function as much more than a thermal fuse, being able to report information regarding the sequence of a series of thermal events but its function is limited to reporting such information after the fact, unlike a computer-monitored thermal couple which may provide live information [1]. However, as no wires are necessary for the operation of the device, large structures may be easily outfitted with multiple sensors. The sensor described herein is capable of operating in extreme environments and, by virtue of its ability to report the important details of its thermal history, allows for failures to be avoided, safety to be improved, and savings in component maintenance to be realized.

2. Concept development

The objective of the sensor developed in this work was to provide a simple procedure by which the thermal history of an engineering component can be queried at intermittent intervals during service. While the microstructure of the material itself is an indicator of thermal history, a more simple approach may be to add or attach something to the material/component which can be queried after the thermal exposure to extract the history. An example of something that might serve in this capacity is a material which undergoes an allotropic phase transformation. Such a material undergoes a change in crystal structure in response to temperature, strain, etc., which results in a change in physical

properties; these changes may be detected by various measurements which can then be correlated back to the temperature, strain, etc., that brought them about. A number of materials are known to undergo allotropic phase transformations as a result of exposure to high temperature; the rate at which such transformations occur (in either the forward or reverse direction) is strongly temperature-dependent. Consequently, by using any one particular phase-transforming material, the measured change in physical properties due to the fraction of material transformed may not correlate in a one-to-one fashion with the transformation-inducing thermal exposure. That is, a short time at high temperature may result in an equivalent amount of transformation as along time at low temperature.

As mentioned above, no single-phase transformation material will provide the necessary thermal history recording function due to the non-uniqueness of a particular degree of transformation. A solution to this problem is to use an array of different phase transformation materials which each transform at different rates and within different temperature ranges. Upon attachment to an engineering component and subsequent thermal exposure, each substrate of the array of phase transformation materials will have transformed to a certain extent which can be measured using any of a number of available means. The resulting pattern of transformations is a "thermal history fingerprint" which corresponds to that particular thermal exposure.

The recording of thermal history fingerprints is a straightforward procedure; complications arise in identifying the thermal exposure that leads to a particular thermal history fingerprint. One solution is to create a large database of known patterns and then use a pattern-matching algorithm for the identifica-

tion of unknown patterns. This scheme for the identification of unknown patterns has been successfully utilized in both law enforcement fingerprint databases as well as the Powder Diffraction File used for the identification of X-ray diffraction patterns of crystalline materials. Both identification methods rely on the existence of a large database of known patterns against which unknown patterns may be compared for identification. In the case of the Powder Diffraction File, there exists the possibility to use computer models to predict patterns; this has the effect of facilitating the identification of unknown patterns by increasing the size of the database against which unknown patterns may be compared. A similar approach may be adopted in identifying the thermal history fingerprints generated by the sensor developed in this work provided an appropriate model can be developed.

A particular class of phase-transformation materials which are especially well suited to this application are glass-ceramic materials [2–5]. Glass-ceramics are alkali, alkaline-earth, or transition metal aluminosilicates with traces of other elements, e.g. TiO_2 , added to improve their transformation behavior by seeding nucleation sites. Glass-ceramic materials are processed by melting the component oxides at high temperature (1200–1600 °C); the melts are then quenched to temperatures in the vicinity of room temperature to recover a clear glass. Upon subsequent heating to the appropriate temperature ranges, nucleation (500–1000 °C) and growth (700–1300 °C) of crystallites within the glass occurs. The typical processing schedule and temperature dependencies of the nucleation and growth rates for these materials is illustrated in Fig. 2. As a result of the crystallization processes, the optical properties of the glass change due to the scattering of light by the crystallites; this is manifested as a decrease in transmittance of the materials with increasing time and temperature of exposure. The transformation of the material from clear glass to opaque polycrystalline aggregate may be easily monitored via a transmittance measurement on substrates of constant dimensions. Additionally, if the crystallizing glass-ceramic materials are heated to temperatures approaching the melting points of the compositions, the crystallites will begin to dissolve back into the glassy phase resulting in less light scattering and an increase in transmittance.

The manner in which the sensor utilizing glass-ceramic materials would actually function is perhaps best illustrated by way of example. A recently developed chemical sensor, referred to as an “electronic nose,” relies on many of the same concepts for its operation [6,7]. The electronic nose consists of an array of different polymeric substrates, each containing a percolating array of carbon black particles. Upon chemical exposure, each polymeric material in the array swells to a different extent; the swelling has the effect of decreasing the connectivity of the percolating carbon particles resulting in an increase in electrical resistance. By measuring the electrical resistance of each polymeric substrate in the array following chemical exposure, a pattern of electrical resistances is obtained that corresponds to the particular chemical. Working in this manner, a database of electrical resistance patterns can be constructed, each corresponding to different chemical exposure. This database can then be used in conjunction with a pattern-matching algorithm to identify unknown resistance patterns and the chemical exposure to which they correspond. The current sensor functions in an analogous manner to the electronic nose with the exceptions that (1) instead of an array of polymeric materials, the sensor employs an array of glass ceramic materials; (2) instead of chemical exposure, the sensor is thermally exposed; (3) instead of measuring electrical resistance, transmittance of the glass-ceramic substrates comprising the sensor is measured. Timely construction of a database of transmittance patterns for a particular application would require some measure of laboratory experimentation under carefully controlled conditions as well as computer modeling.

3. Experimental

3.1. Proof of concept via mini-sensors

Four glass-ceramic compositions from Corning were obtained. The compositions were known to crystallize to cordierite ($\text{Mg}_2\text{Al}_4\text{Si}_5\text{O}_{18}$), canasite ($\text{Na}_4\text{K}_2\text{Ca}_5\text{Si}_{12}\text{O}_{30}\text{F}_4$), β -eucryptite (LiAlSiO_4), and β -spodumene ($\text{LiAlSi}_2\text{O}_6$) within a temperature range of 600–1000 °C [8]. Small substrates

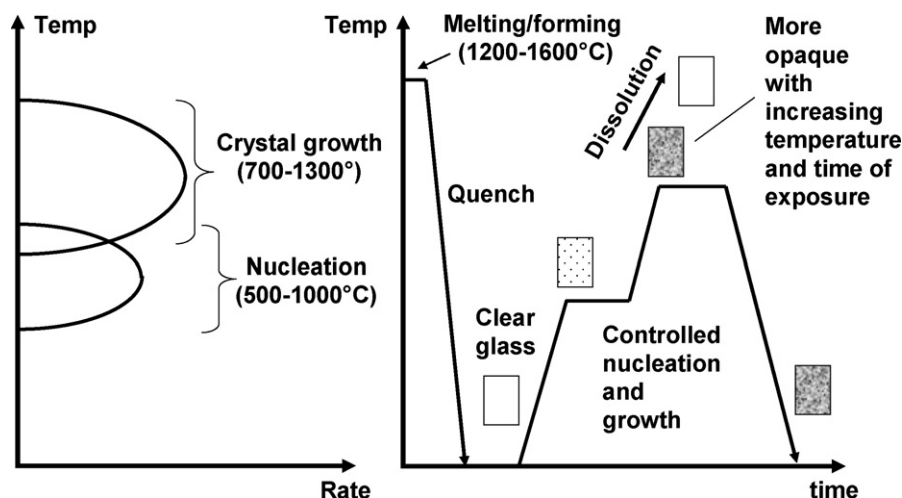


Fig. 2. Typical processing of glass-ceramic materials showing temperature dependence of transformation rates as well as temperature profile during processing.

(5 mm × 5 mm × 2 mm) were machined from each composition and polished on opposing faces to a 1 μm finish using diamond-lapping films. Mini-sensors, comprised of one polished substrate from each composition, were used for heat treatment experiments. Mini-sensors were heat treated at temperatures between 700 and 900 °C for times up to 50 min by placing the sensor in a cordierite boat and sliding the boat in and out of a tube furnace operating at the desired temperature.

Following heat treatment, transmittance of each substrate in the mini-sensor was measured using a UV–vis light spectrophotometer equipped with a photo-diode array detector (Hewlett-Packard Model 8543); this instrument allowed for transmittance measurements across the visible light spectrum to be made in ~5 s. The transmittance of each substrate in the mini-sensor was measured four times, rotating the substrate 90° between each measurement, at a wavelength of 680 nm; the four measurements were then averaged. For a perfectly opaque material, the transmittance measured by the instrument reaches its lowest value; it was assumed that this condition corresponded to the case that the sample was fully crystallized; this condition was used to calculate the extinction coefficient for incorporation into the computer model described below.

3.2. Virtual experiments

A computer model for the crystallization of an array of different glass-ceramic compositions in response to thermal exposure was developed and used to explore the response of a much larger array than that studied experimentally; the model also allowed for the timely investigation of the sensor response to thermal exposures much more complicated than those most easily performed in the laboratory. The model provides a convenient means to illustrate how a transmittance pattern develops as the glass-ceramics in the array crystallize in response to thermal exposure; however, the model cannot be used in reverse to determine thermal histories from a particular transmittance pattern. As stated previously, identification of thermal history will likely require pattern matching using a database of known patterns against which unknowns can be compared. The model takes into account the temperature dependencies of the viscosity, nucleation rate, and growth rate of the glass-ceramic materials. These dependencies were then used in a numerical solution to the Avrami equation for phase transformation to calculate the fraction of material crystallized in each substrate in response to a given heating schedule. The fraction crystallized in each substrate was then used to calculate the transmittance of the substrate; the model then reports the pattern of transmittances for the array as color-coded squares. The modeling employed here is a simplified treatment of the crystallization of glass-ceramic materials but is expected to be realistic enough to paint a semi-quantitative picture of the sensor response.

The temperature dependence of the viscosity, $\eta(T)$, of the i th glass-ceramic composition in the virtual array was modeled using the Volger–Fulcher relation:

$$\eta_i(T) = A_i \exp \left(\frac{Q_i}{T - \tau_i} \right) \quad (1)$$

where T is the temperature in K and, A , Q , and τ are empirical constants for a particular composition [9].

The temperature dependence of the nucleation rate, $I(T)$, of the i th glass-ceramic composition in the virtual array was modeled using the expression suggested by Uhlmann and Yinnon [10]:

$$I_i(T) = \frac{K}{\eta_i(T)} \exp \left[\frac{-0.205 B_i}{(1 - (T/T_{m_i}))^2 (T/T_{m_i})^3} \right] \quad (2)$$

where K is a constant equal to 10^{30} P/cm³ s, B the energetic barrier to nucleation in units of kT at $T = 0.8 T_m$, where T_m is the melting point of the composition and k is Boltzmann's constant. This model assumes homogenous nucleation at a constant rate for a given temperature.

The temperature dependence of the growth rate of the nuclei, $u_i(T)$, of the i th glass-ceramic composition in the array was modeled using the expression

$$u_i(T) = a_0 v_i \left[1 - \exp \left(\frac{-\Delta H_{m_i} (T_{m_i} - T)}{R T T_{m_i}} \right) \right] \quad (3)$$

where a_0 is the diameter of the rate-limiting species (assumed to be on the order of 10^{-10} m), ΔH_m the enthalpy of fusion, and R is the gas constant [10]. The frequency factor v of the i th glass-ceramic composition is given by

$$v_i = \frac{10kT}{3\pi\eta_i(T)a_0^3} \quad (4)$$

This model for growth of the nuclei in complex oxide liquids assumes normal growth of an atomically rough interface.

Using these expressions, a range of transformation rates was achieved by assigning a unique set of constants governing the crystallization behavior (A , Q , τ , B , T_m , and ΔH_m) to each substrate in the model. The constants were chosen such that the resulting temperature-dependencies of viscosity closely approximated those of real glass ceramic materials reported in the open literature; the remaining constants were assigned values lying within the range of experimentally determined values for this class of materials [3,4,8,9,11–15]. The melting temperature, T_m , of each substrate in the array was chosen to be that at which the viscosity was equal to 100 P. The values of the constants used in the model for each composition is listed in Table 1.

Next, a heating schedule, $T(t)$, for the virtual sensor is prescribed and divided into N discrete time intervals of length Δt . The temperature at the j th interval is then given by

$$T(t_j) = T(j\Delta t) \quad (5)$$

The fraction of material crystallized in the i th substrate at the n th time interval is given by [16]

$$x_i(n) = 1 - \exp \left[-\frac{4\pi}{3} \sum_{j=1}^n \left(\sum_{k=j}^n u_i(T(t_k)) \Delta t \right)^3 I_i(T(t_j)) \Delta t \right] \quad (6)$$

This expression represents a numerical integration of the Johnson–Mehl–Avrami equation with the assumption that

Table 1

Values of constants used to model individual glass-ceramic compositions in sensor model

<i>i</i>	<i>A</i> (P)	<i>Q</i>	τ	<i>B</i>	<i>T_m</i> (°K)	ΔH_m (kJ/mol)
1	10 ⁻¹⁰	39,000	0	105	1411	280
2	10 ⁻¹⁰	40,000	0	104	1448	360
3	10 ⁻¹⁰	41,000	0	103	1484	320
4	10 ⁻¹⁰	42,000	0	102	1520	200
5	10 ⁻¹⁰	43,000	0	101	1556	280
6	10 ⁻¹⁰	44,000	0	100	1592	320
7	10 ⁻¹⁰	45,000	0	95	1629	200
8	10 ⁻¹⁰	46,000	0	94	1665	280
9	10 ⁻¹⁰	47,000	0	93	1701	80
10	10 ⁻¹⁰	48,000	0	92	1737	240
11	10 ⁻¹⁰	49,000	0	91	1773	200
12	10 ⁻¹⁰	50,000	0	90	1809	240
13	10 ⁻¹¹	39,000	-40	100	1499	320
14	10 ⁻¹¹	41,000	-80	100	1539	360
15	10 ⁻¹¹	43,000	-120	100	1578	200
16	10 ⁻¹¹	45,000	-160	100	1616	280
17	10 ⁻¹¹	46,500	-200	100	1636	160
18	10 ⁻¹¹	48,000	-240	100	1655	200
19	10 ⁻¹¹	51,000	-300	100	1713	120
20	10 ⁻¹¹	53,000	-340	100	1752	200
21	10 ⁻¹¹	55,000	-380	100	1791	240
22	10 ⁻¹¹	57,000	-420	100	1830	160
23	10 ⁻¹¹	58,500	-460	100	1849	320
24	10 ⁻¹¹	60,000	-500	100	1868	360
25	10 ⁻¹¹	59,000	-460	86	1850	280
26	10 ⁻¹¹	58,000	-420	86	1850	200
27	10 ⁻¹¹	57,000	-380	87	1850	120
28	10 ⁻¹¹	56,000	-320	87	1850	80
29	10 ⁻¹¹	54,000	-280	88	1850	240
30	10 ⁻¹¹	53,000	-240	88	1850	200
31	10 ⁻¹¹	52,000	-200	89	1850	160
32	10 ⁻¹¹	51,000	-160	89	1850	320
33	10 ⁻¹¹	49,500	-120	90	1850	360
34	10 ⁻¹¹	48,000	-80	90	1850	280
35	10 ⁻¹¹	47,000	-40	91	1850	200
36	10 ⁻¹¹	46,000	0	91	1850	320

growth of the nuclei is spherical; this model also takes into account impingement of crystallized regions in the latter stages of the transformation [16–18]. The integration was performed following each time step, Δt , in the virtual experiment; time steps of 20 s were used for all experiments.

For the purposes of relating the fraction crystallized as predicted by the model to the observed transmittance in a simple and straightforward manner, the measured transmittance, T , was assumed to follow a Beer's law relation [19]:

$$T = \exp[-abx] \quad (7)$$

where a is the specimen thickness, b the instrument-dependent extinction coefficient, and x is the fraction of crystallized phase in the material. In this case, Beer's law is an oversimplification and fails to take into account a number of processes that may affect the transmittance such as reflection, absorption, and the effect of particle size on light scattering; however, for the purposes of illustrating that the transmittance is proportional to the fraction of materials crystallized, i.e. that which scatters light, and the thickness of the specimen, Beer's law will suffice. The

corresponding transmittance of the i th substrate is given by

$$T_i(n) = \exp(-kx_i(n)) \quad (8)$$

where k is a constant accounting for the specimen thickness as well as the extinction coefficient of the measurement instrument. The constant k was chosen to match that obtained measuring the transmittance of the glass-ceramic substrates used in the mini-sensor experiments: $k = 10$.

The model then returns the transmittance of each substrate in the 6×6 array at the n th time interval as a matrix:

$$T_{6 \times 6} = \begin{bmatrix} T_1(n) & T_2(n) & T_3(n) & T_4(n) & T_5(n) & T_6(n) \\ T_7(n) & T_8(n) & T_9(n) & T_{10}(n) & T_{11}(n) & T_{12}(n) \\ T_{13}(n) & T_{14}(n) & T_{15}(n) & T_{16}(n) & T_{17}(n) & T_{18}(n) \\ T_{19}(n) & T_{20}(n) & T_{21}(n) & T_{22}(n) & T_{23}(n) & T_{24}(n) \\ T_{25}(n) & T_{26}(n) & T_{27}(n) & T_{28}(n) & T_{29}(n) & T_{30}(n) \\ T_{31}(n) & T_{32}(n) & T_{33}(n) & T_{34}(n) & T_{35}(n) & T_{36}(n) \end{bmatrix} \quad (9)$$

The transmittance of the i th substrate in the array is a number between 1 (full transmittance) and 0 (opaque); for clarity of presentation, the transmittances reported in the matrix may be color coded in grey scale from white ($T = 1$) to black ($T = 0$). For the purposes of calibration of the grey scale, the matrix elements $T_1(n)$ and $T_{36}(n)$ were set to 0 and 1, respectively; this has the same effect as using non-crystallizing or fully opaque reference elements in the actual sensor. Consistent with the fact that the measured transmittances from the mini-sensor experiments had an error of approximately $\pm 3\%$, the grey scale used for color-coding was divided into 20 bins ($T = 0-0.05$, $T = 0.06-0.10$, etc.) in order to approximate the coarseness of the experimental measurement.

3.3. Experimental confirmation of model predictions using larger sensor

To confirm that the predictions of the semi-quantitative model could in fact be realized in real life, experiments were performed using a larger sensor comprised of the four glass-ceramic materials described previously as well as an additional four compositions for a total of eight different compositions. The additional four compositions were obtained as custom melts from Corning with compositions in the Mg–Al–Si–O, Ca–Al–Si–O, Ba–Si–Al–O, and Na–Ba–Si–Al–O ternary and quaternary systems. The exact crystallization behavior (crystallization rates, resulting crystalline phases, etc.) was not studied nor is it critical to fully understand such behavior to utilize the materials as sensing elements in the thermal history sensors under consideration; rather, it is more important that the material be of sufficient homogeneity such that the crystallization behavior for a given thermal exposure is always the same. Thermal sensors comprised of small substrates of the eight glass-ceramic compositions were loaded in to a cordierite boat and placed in a three-zone tube furnace with temperatures of 600, 800, and 900 °C. The boat was then moved in and out of the different

temperature zones to yield two different thermal cycles; cycle A was comprised of 10 min at 600 °C followed by 10 min at 800 °C while cycle B was comprised of 10 min at 600 °C followed by 10 min at 900 °C. In each experiment, the sensor was exposed to four such cycles, each either A or B, leading to 16 different thermal histories, e.g. AAAA, ABBA, BBBA, etc. The transmittances of the sensing elements of each sensor were then measured at 680 nm as described earlier in this section.

4. Results

4.1. Proof of concept via mini-sensors

A photograph of the results of the mini-sensor heat treatment experiments is shown in Fig. 3. The substrates are backlit from beneath and each is labeled with ink to identify the crystallizing phase: I for β -spodumene, Co for cordierite, E for β -eucryptite, and Ca for canasite; the ink also serves to illustrate the change from clear to translucent and opaque for the materials. It is clear from the photograph that the optical properties of the glass-ceramic materials do indeed change as a function of thermal exposure. For example, the composition crystallizing to canasite is shown to become translucent and then opaque as a consequence of increasing time and temperature of exposure. Similarly, the composition crystallizing to β -spodumene is shown to darken and then opacify as a consequence of increasing time and temperature of exposure. The measured transmittances of the substrates comprising the mini sensors is illustrated in Fig. 3; the measured transmittances have been assigned a grey scale value linearly ranging from white for fully transmissive (normalized transmittance $T/T_0 = 1$) to black for fully opaque ($T/T_0 = 0$). The extinction coefficient, b , of the spectrophotometer was determined to be $\sim 5000 \text{ m}^{-1}$.

4.2. Virtual experiments

Fig. 4 shows the results of virtual experiments using the computer model for the case of cyclic heating. The virtual sensor was

exposed to cycling heating between 500 and 950 °C with a saw tooth pattern and a period of 80 s. The figure shows the transmittance patterns obtained for the indicated number of cycles. It can be seen that a different pattern of transmittances is obtained depending on the number of cycles.

Fig. 5 shows the results of virtual experiments using the computer model for the case of cyclic heating (25 total cycles) with one anomalous cycle. The cyclic heating is the same as that shown in the previous figure and the anomalous cycle in this case is a +150 °C overheat (up to 1100 °C). The figure shows the transmittance pattern for the case of 25 cycles with no overheats as well as the patterns obtained for one +150 °C overheat at the 2nd, 10th, and 20th cycle. Inspection of the transmittance patterns in this case reveals that distinct transmittance patterns are obtained depending on (1) whether or not an overheat has occurred and (2) if so, when the overheat has occurred.

Fig. 6 shows the results of virtual experiments using the computer model for the case of cyclic heating (100 total cycles) with multiple anomalous cycles. The cyclic heating is the same as that shown in the previous figures and the anomalous cycles are overheats between +100 and +180 °C. As shown in the figure, transmittance patterns which appear very similar can be obtained for thermal histories which are quite distinct.

4.3. Experimental confirmation of model predictions using larger sensor

Fig. 7 shows the temperatures and times associated with the two thermal cycles employed for these experiments. Fig. 7 also shows photographs of the exposed sensors as well as patterns in which the transmittances have been color coded to grey scale. For the purposes of correctly adjusting the contrast of the photographic images, additional reference elements were included. These reference elements appear at either end of the sensor array (black-opaque and white-fully transmissive); an additional partially crystallized element was used to set the middle grey (50% transmission) and appears in the images next to the white refer-

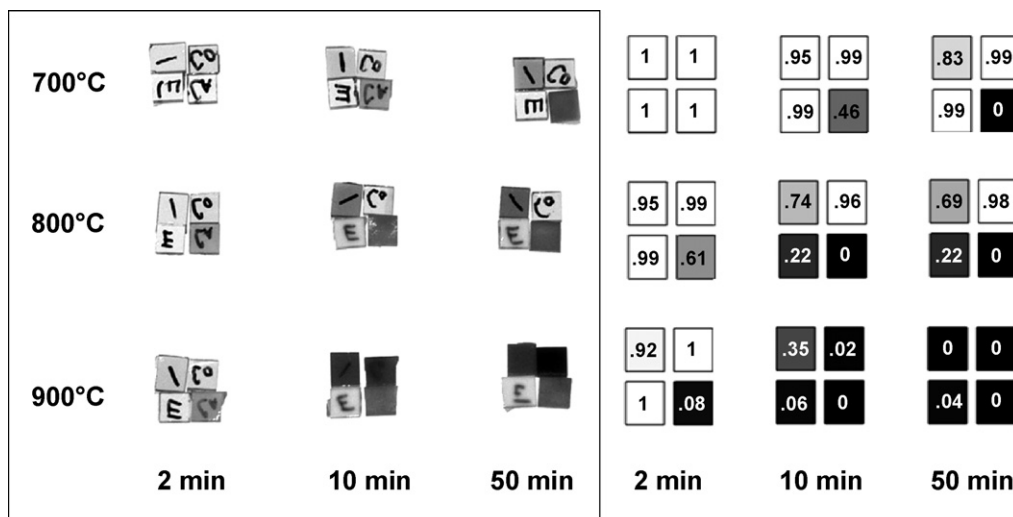


Fig. 3. Photograph of thermally exposed mini-sensors along with figure of depicting measured transmittances in the greyscale output used by the model. Legend—I: spodumene; Co: cordierite; E: eucryptite; Ca: canasite.

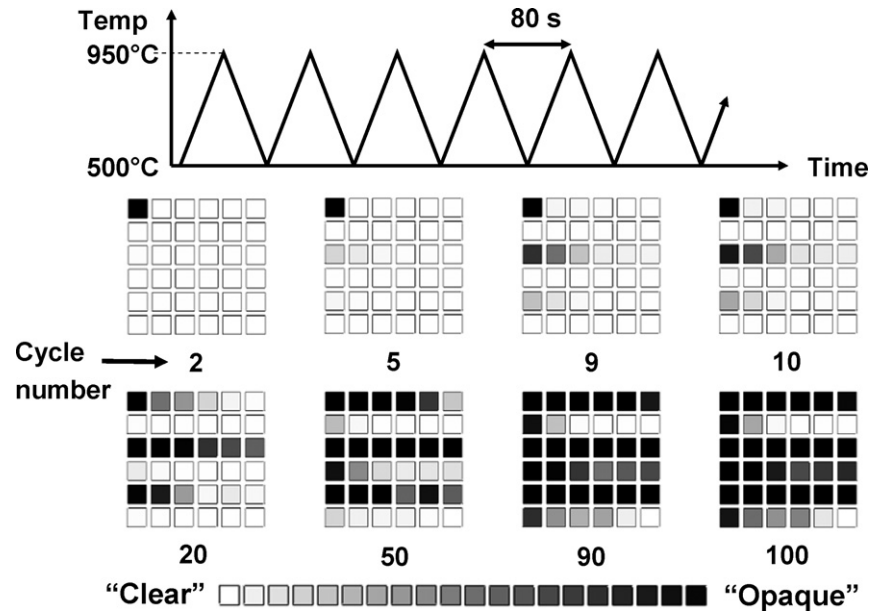


Fig. 4. Model output showing sensor response due to cyclic heating.

ence element with a black dot near the top. The differences in the photographs of the sensors are difficult to identify; in contrast, the differences in the patterns of color-coded transmittance obtained at a fixed wavelength of light are much easier to discern. It is clear that the transmittance patterns for the thermal histories shown (AAAA, AAAB, AABA, ABAA, BAAA) are distinguishably different. In this case it is clear that the thermal sensor does in fact function as predicted by the computer model: the sensor is able to indicate if and when a B cycle occurs thereby providing information about both the temperatures involved as well as the sequence of thermal events. The measured transmittances are not necessarily consistent with the model; rather it is

the overall behavior of the actual sensor that is consistent with that predicted in our virtual experiments.

5. Discussion

The results of the mini-sensor experiments demonstrate the sensor concept in a simple and straightforward fashion. As shown in Fig. 3, according to the time and temperature of exposure, a unique pattern of transmittances is obtained as a consequence of the varying rates of crystallization in the glass-ceramic materials considered. It is feasible to construct a database of sensor responses, in the form of patterns of transmit-

25 total cycles, +150°C overhear at n^{th} cycle:

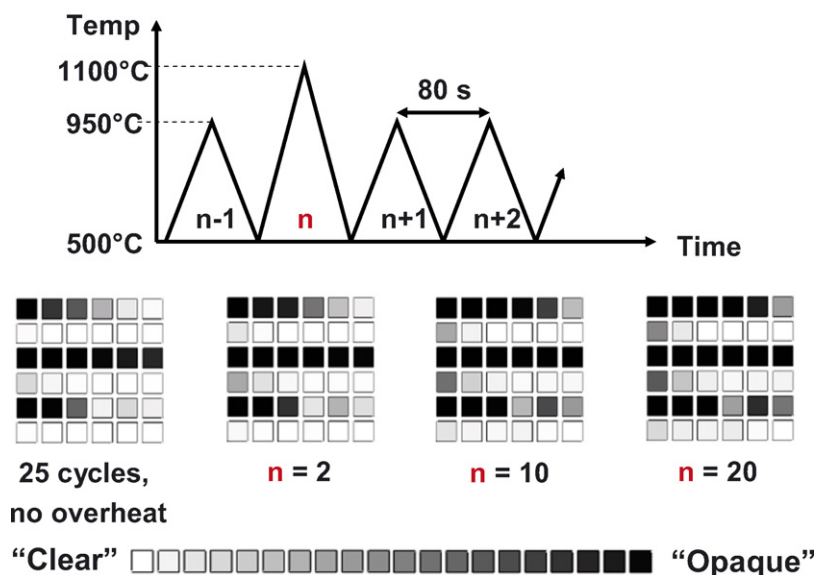


Fig. 5. Model output showing sensor response due to cyclic heating with 150 °C overhear at indicated cycle number.

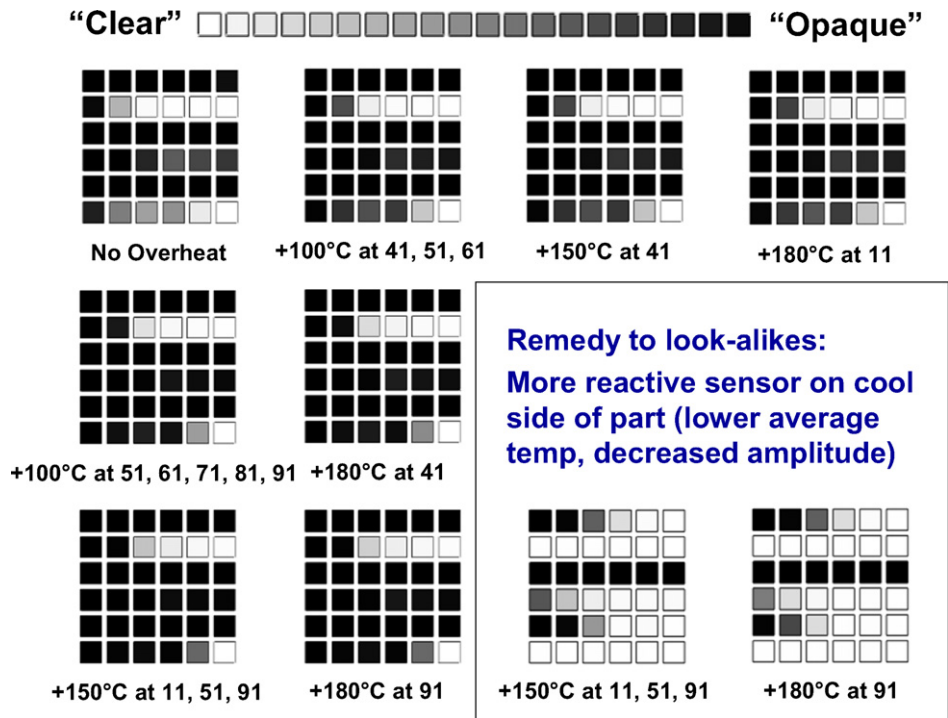


Fig. 6. Model output showing sensor response to 100 total cycles with $X^{\circ}\text{C}$ overheat at the indicated cycle numbers. Similar transmittance patterns may be obtained for differing thermal histories.

tances, for the time–temperature space considered here (single stage heating, $T=700\text{--}900^{\circ}\text{C}$, time = 2–50 min). Any transmittance pattern subsequently obtained as a consequence of one stage heating in this time–temperature space could be identified by comparison to the database, thereby identifying the thermal history responsible for the observed transmittance pattern.

The results obtained in the virtual experiments illustrate the functionality of a larger array sensor in response to more complex thermal histories such as cyclic heating with overheat events. As shown in Fig. 4, a unique pattern of transmittances

is obtained which depends on the number of thermal cycles the sensor is exposed to; in these cases, the sensor is able to report information regarding its history that a simple thermal fuse or thermal paint could not, i.e. the number of thermal cycles to a particular temperature. This sensing capability is useful for monitoring the health of components for which thermomechanical fatigue leads to failure. Furthermore, as illustrated in Fig. 5, the addition of overheat events to the cyclic heating described above leads to distinct patterns depending on when such events occur during cycling. In these cases, in addition to information

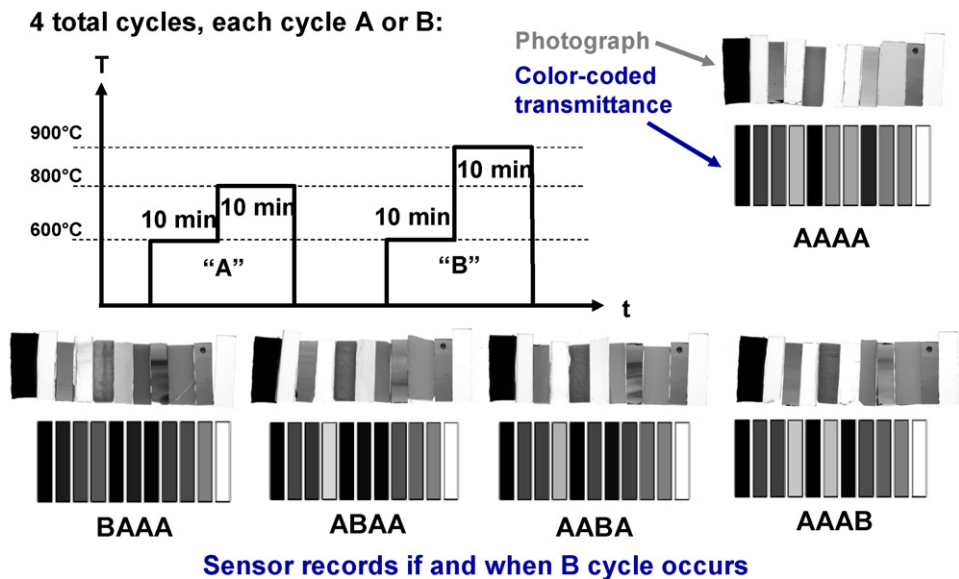


Fig. 7. Thermal cycles used to expose large sensor as well as photographs of the exposed sensors and color-coded transmittance patterns.

regarding the total number of cycles, the sensor is able provide information regarding the sequence of events, i.e. whether the overheat events occurred early on or later during cycling. This extends the applicability of the sensor to monitoring components subject to thermomechanical fatigue and also limited in lifetime by their maximum temperature of exposure. The information reported by the sensor in these instances becomes an abbreviated version of that provided by a continuously monitored thermal couple, i.e. a rough account what happened and when it happened. The fact that such behavior was also realized in laboratory experiments with the large sensor demonstrates the viability of this approach to thermal sensing.

During the course of virtual experimentation, it was discovered that different thermal histories could in fact lead to transmittance patterns that appear very similar. The problem of obtaining similar transmittance patterns for differing thermal histories illustrated in Fig. 6 can be addressed through the use of multiple sensors or by relocating sensors on the component of interest. Fig. 6 illustrates this idea for the case in which the virtual sensor was relocated to a different location on the virtual component; in this case, the sensor was relocated to a cooler region of the component which is easily simulated by reducing the average temperature and amplitude of thermal cycling. As a consequence of the relocation, Fig. 6 shows that the previously similar transmittance patterns are now distinct.

An alternative approach is to consider that obtaining similar transmittance patterns for differing thermal histories poses no problem as long as the microstructural degradation in the component corresponding to the thermal histories in question is similar as well. For these cases, the thermal history sensor would actually function as a degradation-state sensor; from the point of view of anticipating and preventing failures of engineering components, this function is equally useful as that of reporting a thermal history. This concept was briefly explored in the current work using expression for the temperature dependence of degradation mechanism for real engineering materials reported in the literature. For example, by fitting the data from the deformation-mechanism map of Frost and Ashby, the temperature dependence of the creep rate of nickel with a grain size of 10 μm at an applied stress of $\sigma/\mu = 10^{-4}$ is given by [20]

$$\dot{\epsilon}(T) \approx 6.04 \exp\left(\frac{-14300}{T}\right) (\text{s}^{-1}) \quad (10)$$

This expression can be numerically integrated with respect to time assuming particular thermal history, $T(t)$, to obtain the total creep strain:

$$\epsilon = \sum_j^n \dot{\epsilon}(T(t_j)) \Delta t \quad (11)$$

The total creep strain obtained by the integration for the thermal histories shown in Fig. 6, which lead to similar transmittance patterns, is reported in Table 2. It can be seen that thermal histories which result in similar transmittance patterns do correspond to similar amounts of creep strain.

Similarly, by fitting the data reported by Ardell, the temperature dependence of the coarsening rate of Ni_3Al precipitates in

Table 2

Integrated damage to thermal histories leading to similar transmittance patterns

Thermal history	Integrated creep strain	Integrated grain size (nm)
100 cycles, no overheat	0.0832	53.5
100 cycles, +100 °C at 41, 51, 61	0.0871	56.4
100 cycles, +150 °C at 41	0.0868	56.0
100 cycles, +180 °C at 11	0.0879	57.5

Ni–Al alloys is given by [21]

$$K(T) \approx 10^{13} \exp\left(\frac{-32000}{T}\right) (\text{nm}^3/\text{s}) \quad (12)$$

With the assumption that this expression holds over the entire Ni– Ni_3Al phase field [22], this expression can be numerically integrated as in the previous example to provide a measure of the grain size:

$$r \approx \sqrt[3]{\sum_j^n K(T(t_j)) \Delta t} \quad (13)$$

The final grain size obtained by the integration for the same thermal histories used in the last example is reported in Table 2. As in the previous example, thermal histories resulting in similar transmittance patterns also result in similar coarsened grain sizes.

Despite the sensor's ability to record information regarding thermal history, there are some potential limitations to its operation. One potential limitation is the level of detail of the thermal history that the sensor can report. For instance, when subjected to thermal cycling with overheat events, the sensor may not register the overheat events if they are not of sufficient magnitude. For the model considered here, the overheat event must be greater than $\sim 50^\circ\text{C}$ to be detected. Similarly, for a large number of cycles, the sensor may not be able to distinguish between an overheat that occurred at, say, the sixth or seventh cycle of 100 total cycles; rather, the sensor may only be able to report that an overheat did occur in the time frame of the first 10 cycles. Due to the semi-quantitative character of the model, these features of the sensor are expected to be manifest in real glass-ceramic materials as well. Determination of the sensitivity of the sensor as well as the limits of its recording abilities are the subject of ongoing experimentation.

Another potential limitation is the effect of environmental exposure on the operation of the sensor. The viscosity of silicate glasses decreases nonlinearly with increasing water content with possible decreases on the order two degrees of magnitude at a given temperature [23]. With the nucleation and growth rates of glass-ceramic materials being inversely dependent on the viscosity of the glass, dramatic changes in the crystallization behavior are to be expected due to moisture in the environment. The effect of moisture on the crystallization of glass-ceramic materials may complicate the identification of the transmittance patterns obtained for similar thermal histories but different environmental moisture contents. For engineering components

which operate at high temperature in a variety of moisture environments, i.e. aircraft engine exhaust components, the effect of moisture may prove an insurmountable barrier to the implementation of the sensor; a means to encapsulate the sensor from the environment may prove necessary. For components which see high temperatures during use in a relatively constant moisture environment, i.e. refractory materials processing equipment, the sensor may also function as a moisture or environmental indicator in that deviations from transmittance patterns known for a particular process may be traced to the fluctuation of environmental variables which could negatively impact the process.

6. Conclusions

The experiments performed in the current work, both real and virtual, have demonstrated a thermal sensor which records information about the thermal history as a pattern of transmittances in an array of glass-ceramic materials. The sensor is able to function in a larger capacity than a thermal fuse, being able to report information regarding the number of thermal excursions and sequence of thermal events; at the same time, the sensor functions at a level below that of a computer-monitored thermal couple in that only a coarse measure of the thermal history is provided. The sensor also has the advantages of being wireless (similar to a thermal fuse) and obviating the need for data storage (in contrast to a computer-monitored thermal couple) as the information regarding thermal history is physically contained in the sensor and can be extracted as needed. In addition to thermal sensing, the sensor may also provide a measure of the degradation state of the monitored materials and components. The sensor is expected to be useful in a number of extreme environment applications, subject to the potential limitations described in the text, in which the failure of engineering components due to thermal exposure is to be avoided.

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